4.1 Practice Problems

Name___________________________________

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Approximate the number using a calculator. Round your answer to three decimal places.

1) $5^{1.9}$ 1) ________________

2) $5^{-1.9}$ 2) ________________

3) $4^{\sqrt{6}}$ 3) ________________

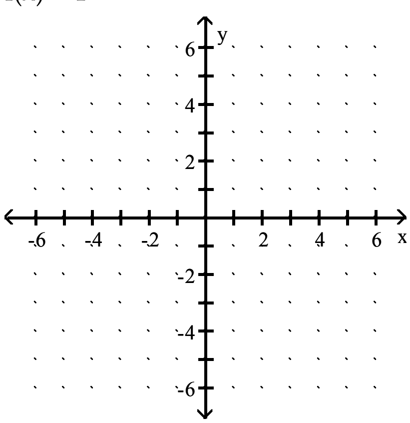
4) 3.6^{π} 4) ________________

Solve the problem.

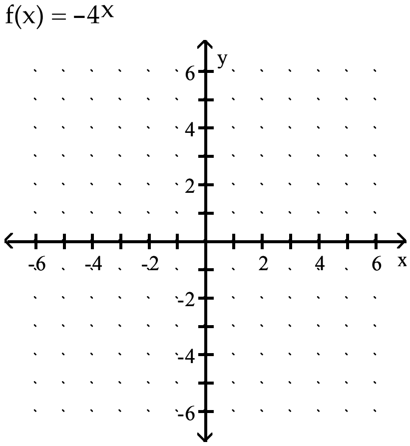
5) A city is growing at the rate of 0.6% annually. If there were 3,812,000 residents in the city in 1994, find how many (to the nearest ten–thousand) are living in that city in 2000. Use $y = 3{,}812{,}000(2.7)^{0.006t}$. 5) ________________

Graph the function by making a table of coordinates.

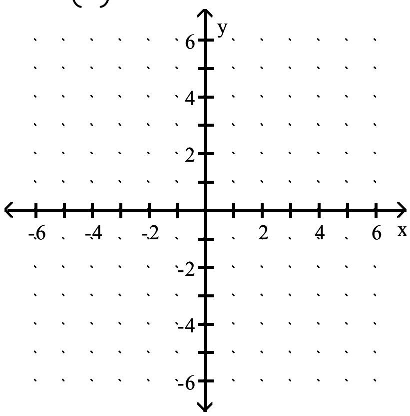
6) $f(x) = 4^x$ 6) ________________



7) $f(x) = -4^x$ 7) ________________



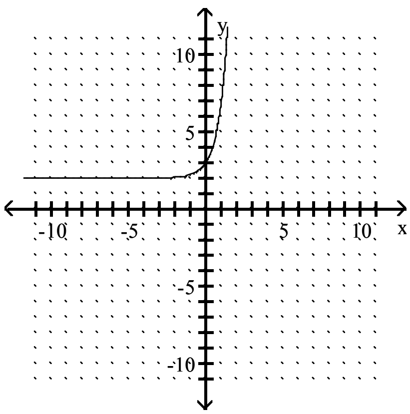
8) $f(x) = \left(\frac{1}{4}\right)^x$ 8) ________



MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

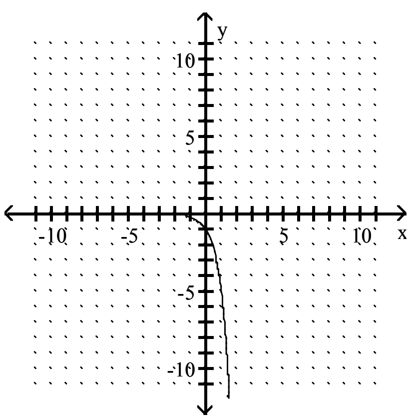
The graph of an exponential function is given. Select the function for the graph from the functions listed.

9) 9) ________



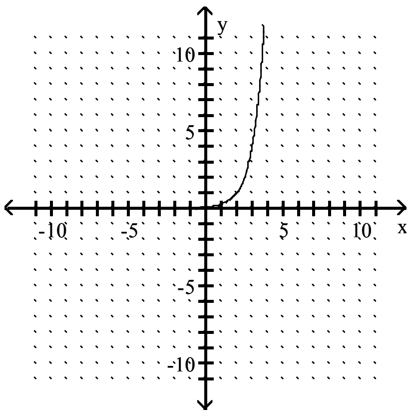
A) $f(x) = 5^{x+2}$ B) $f(x) = 5^x - 2$ C) $f(x) = 5^x + 2$ D) $f(x) = 5^x$

10) 10) ________



A) $f(x) = -5^{-x}$ B) $f(x) = 5^x$ C) $f(x) = 5^{-x}$ D) $f(x) = -5^x$

11) 11) _______

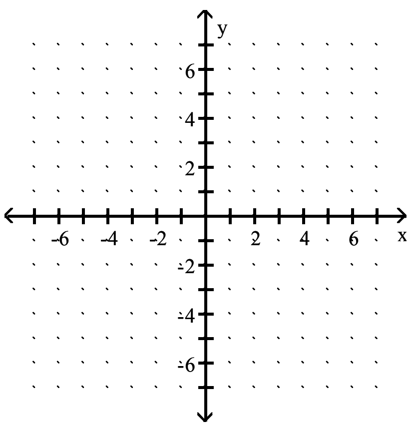


A) $f(x) = 4^x - 2$ B) $f(x) = 4^x + 2$ C) $f(x) = 4^x$ D) $f(x) = 4^{x-2}$

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

Graph the function.

12) Use the graph of $f(x) = e^x$ to obtain the graph of $g(x) = -e^x$. 12) _______________



Approximate the number using a calculator. Round your answer to three decimal places.

13) $e^{1.1}$ 13) _______________

14) $e^{-0.7}$ 14) _______________

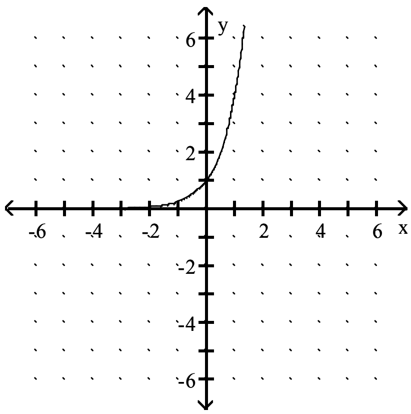
Solve the problem.

15) The size of the coyote population at a national park increases at the rate of 4.6% per year. If the size of the current population is 130, find how many coyotes there should be in 6 years. Use the function $f(x) = 130e^{0.046t}$ and round to the nearest whole number. 15) _______________

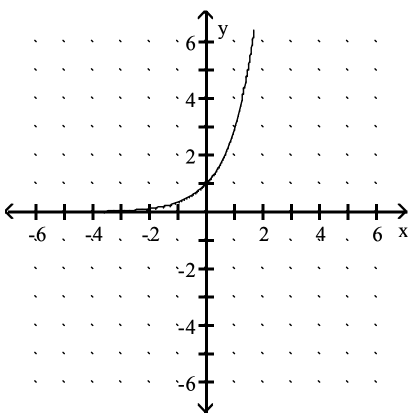
Use the compound interest formulas $A = P\left(1 + \frac{r}{n}\right)^{nt}$ and $A = Pe^{rt}$ to solve.

16) Find the accumulated value of an investment of $19,000 at 12% compounded annually for 13 years. 16) _______________

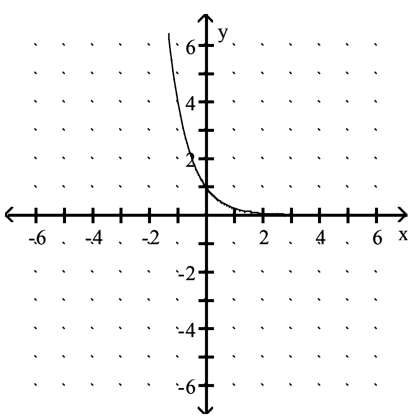
1) 21.283
2) 0.047
3) 29.836
4) 55.934
5) 3,950,000
6)



7)

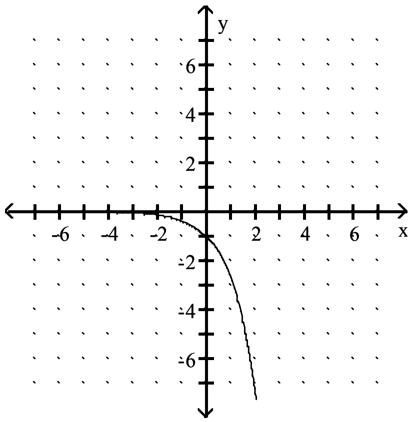


8)



9) C
10) D
11) D

12)



13) 3.004
14) 0.497
15) 171
16) $82,906.37